

* Theory

1) What is Inferential Statistics?

⇒ Inferential statistics is a branch of statistics that uses sample data to draw conclusions or make predictions about a large population.

2) What is Hypothesis Testing and its components?

⇒ It is a statistical method used to determine whether there is enough evidence to support a claim about a population.

* The components are:-

⇒ Null Hypothesis (H_0)

⇒ Alternative Hypothesis (H_1)

⇒ Significance Level (α)

⇒ Test statistic

⇒ P-value

⇒ Decision Rule

3) Explain confidence Interval and Critical Value.

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* confidence interval:-

It is a Range of values that is likely to contain the true population parameter with a specified level of confidence, It is usually 95%.

* Critical value:

A critical value is the threshold used to decide whether to reject the null hypothesis.

W Define P-value:

The p-value is the probability of obtaining the observed sample results, assuming that the null hypothesis is true.

- If $p\text{-value} < 0.05$, reject the null hypothesis.
- If $p\text{-value} \geq 0.05$, fail to reject the null hypothesis.

5) Differentiate Between Type I and Type II errors:

Type I errors	Type II errors
→ Rejecting a true null hypothesis	→ Failing to reject a false null hypothesis.
→ False Positive	→ False Negative
→ Probability of error	→ Probability of error

6] Brief Description of Statistical Tests

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* Z-Test

A Z-Test is used to compare sample size means when the sample size is large or when the population standard deviation is known.

* T-Test

A T-Test compares the means of two independent groups when the sample size is small or the population standard deviation is unknown.

* Chi-square test

The chi-square test determines whether there is a significant association between two categorical variables.

* ANOVA (Analysis of variance)

It compares the mean of three or more groups to determine whether at least one group mean differs significantly from the others.

7) What is covariance?

Covariance measures how two variables change together.

- Positive covariance indicates that both variables increase or decrease together.
- Negative covariance indicates that one variable increases while the other decreases.

8) What is correlation?

It measures the strength and direction of the linear relationship between two variables.

The correlation coefficient ranges from -1 to +1.

- +1 : Perfect positive correlation.
- 0 : No correlation.
- -1 : Perfect negative correlation.

Correlation is widely used in statistical analysis, finance, healthcare, and scientific research.